

CoRoT-2b

R-1732

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130714 · created 2026-08-02T17:44:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456487.7421 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.221 +/- 0.067
Transit depth (Rp/Rs)^2	4.87 +/- 2.95 [%]
Orbital Inclination (inc)	89.07 +/- 2.08
Airmass coefficient 1 (a1)	1.33 +/- 0.9
Airmass coefficient 2 (a2)	-0.13 +/- 0.43
Scatter in the residuals of the lightcurve fit is	76.41 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	5.76
Transit Duration (day)	0.0993 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.221 $\pm$ 0.067 vs 0.1667 (deviation 0.6 σ)
Duration fit vs published	0.0993 d vs 0.095 d (deviation 0.45 σ)
Mid-time O-C	-5.0 min
Depth SNR	2.4
Residual scatter	76.41 %

Light curve

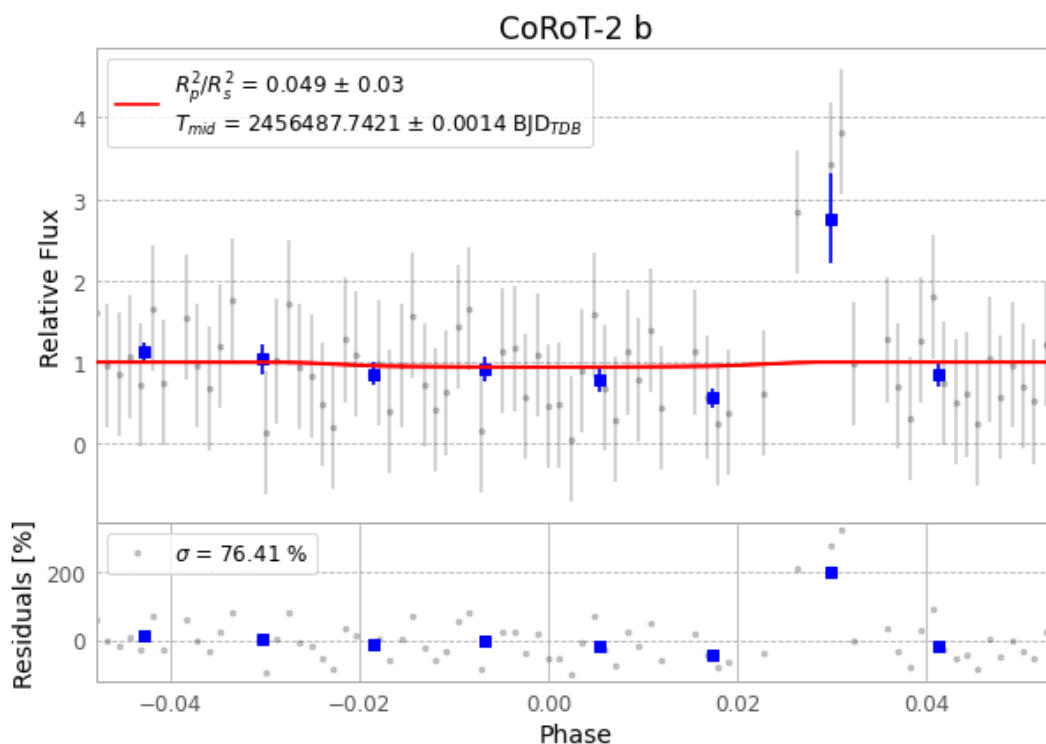


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-07-13.png (SHA-256 833770a85f4cda85...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 753bbd67b9e0bdce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2130714033912.FITS → CoRoT-2130714075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130714.log (815ffd6526d4...), FinalLightCurve_CoRoT-2 b_2013-07-13.png (833770a85f4c...), FinalLightCurve_CoRoT-2 b_2013-07-13.csv (60ef97296593...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:44Z.